

DengAI — Predicting Disease Spread

Metrics and Models

October 2026

Metrics

The competition metric is MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

The average number of cases we are off by, per week.

- One number over all **416 test weeks** (260 San Juan + 156 Iquitos)
- Lower is better; a perfect model scores 0
- **San Juan dominates:** 63% of the weeks and about four times the cases

The target is skewed

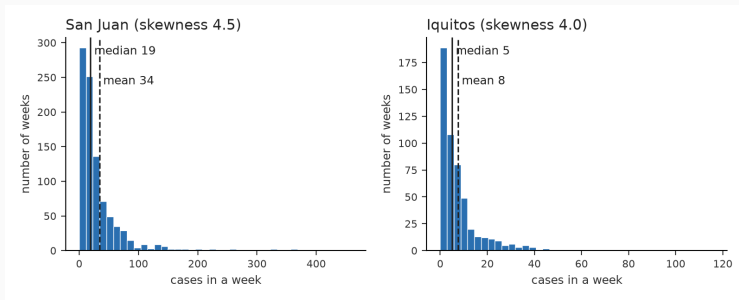


Figure 1: Weekly cases: most weeks are quiet, a few outbreak weeks form a long tail

San Juan: **mean 34, median 19**, 73% of weeks below the mean. Skewness **4.5** (San Juan) and **4.0** (Iquitos).

Why MAE and not RMSE

- RMSE squares errors: one 461-case week counts like **~500 ordinary weeks off by one**
- RMSE is dragged into fitting a few outbreaks; **MAE weights every week equally**
- MAE is minimised by the **median**, RMSE by the **mean** — for skewed counts the median is the robust target
- Consequence: good models here are *conservative*

Models

The pipeline has four branches

`make_city_pipeline` builds one of four shapes, because model families need different preprocessing:

Branch	Steps	Models
tree	impute → history → cyclical → select → model	CatBoost, RandomForest
dense	tree steps + impute + scale	Ridge, Lasso
time_series	impute → history → date + scaled weather	Prophet, SARIMAX
seasonal	calendar columns only	Seasonal median, Shape×level

Why four branches

`dense` — the lag/rolling warm-up leaves **NaNs** in the first 26 weeks

- Tree models handle NaN natively; RidgeCV raises `ValueError`
- So linear models get `SimpleImputer(median) + StandardScaler`

`seasonal` — the calendar baselines need `year` and `weekofyear`, which the tree branch drops

- They **never see the weather**: the baseline any weather model must beat

The nine models

Model	Family	Core assumption
CatBoost	gradient boosting	non-linear effects of lagged weather
XGBoost	gradient boosting	as CatBoost, on RFE-selected features
RandomForest	bagged trees	same, without boosting
Ridge	linear + L2	smooth linear response, all features shrunk
Lasso	linear + L1	linear, but most features are irrelevant
Prophet	additive time series	trend + annual seasonality + regressors
SARIMAX	ARIMA + exog	autocorrelated errors + Fourier seasonality
Seasonal median	calendar only	this week behaves like the same week in past years
Shape $\times$ level	calendar only	seasonal <i>shape</i> $\times$ typical annual <i>total</i>

Results — holdout MAE

Model	San Juan	Iquitos
CatBoost	15.02	6.27
RandomForest	15.16	8.39
XGBoost + RFE	17.21	3.54
Shape × level	20.72	3.39
Lasso	21.16	4.50
Prophet	21.78	7.16
Seasonal median	23.49	4.41
Ridge	24.46	5.48
SARIMAX	24.56	4.61

Different winners per city — the submission uses **CatBoost for San Juan**, **SARIMAX for Iquitos**.

What drives the CatBoost predictions?

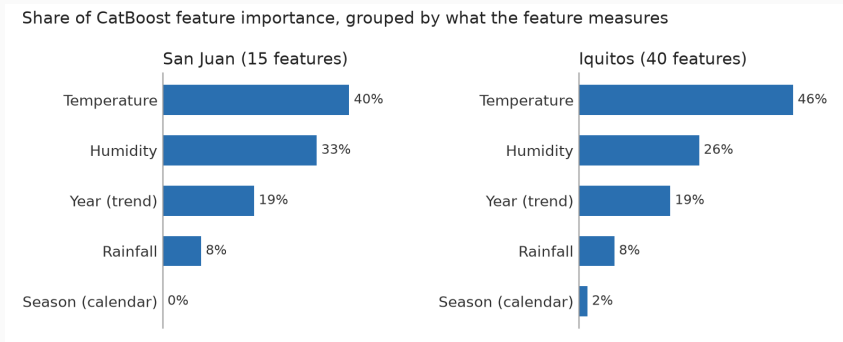


Figure 2: CatBoost feature importance by feature family

- **Temperature and humidity: about 70%** of importance in both cities
- Top features are **8–26-week averages**, not this week's weather
- **year: about 19%** — levels drift between years for non-weather reasons

Why weather is not the whole story

- Climate predicts **when** cases rise within a year, not **how many**
- In Iquitos the calendar-only `Shape × level` (3.39) beats every weather model
- Epidemic size depends on the circulating **serotype** and population immunity — neither is in the data

- **Metrics** — MAE, because the target is skewed (4.5): a few outbreak weeks would dominate RMSE
- **Models** — nine models in four pipeline branches; winners differ by city
- **Feature importance** — months of temperature and humidity; weather says *when*, not *how many*